FactorGCL: A Hypergraph-Based Factor Model with Temporal Residual Contrastive Learning for Stock Returns Prediction Supplemental Materials

Environment

All the experiments are conducted with following settings:

- CPU: Intel Xeon Silver 4210 CPU.
- GPU: Nvidia Tesla P40.
- Operating system: Ubuntu 16.04.7 LTS.
- Software versions: Python 3.7.16; Numpy 1.21.5; Pandas 1.3.5; Pytorch 1.8.1, scipy 1.7.3.

Data Preprocessing

The dataset in our experiments is constructed from a sequence of market data comprising day-level price-volume data (*high*, *open*, *low*, *close*, *volume-weighted average price*, and *trading volume*).

We first look back at the past $T = 60$ days of data to obtain a sequence of 60×6, then we preprocess the data by dimensionless processing. Specifically, for the price dimension data, we divide by the current closing price , to obtain the relative price, and for the trading volume dimension data, we divide by the average trading volume , to obtain the relative trading volume. The construction of future sequence data is similar. For the label, we use the future stock returns with multiple periods ($\Delta t = 1, 5, 10, 20$), and obtain the label after cross-sectional standardization. The formula is as follows:

$$\tilde{y}_t = \frac{\text{price}_{t+\Delta t+1} - \text{price}_{t+1}}{\text{price}_{t+1}}$$
$$y_t = \frac{\tilde{y}_t - \mu_t}{\sigma_t}$$

where $\tilde{y}_t$ is the raw stock return at time t, μ_t and σ_t are the mean and standard deviation of $\tilde{y}_t$ in cross-sectional stocks at time t.

Finally, we drop samples with too many missing values, fill missing values with 0, and clip extreme values in the dataset

Implementation Details

In our experiments, we use the Adam optimizer with a learning rate of 1×10^{-3}. All models are trained for 100 epochs, with early stopping set to 20 steps, and the random seed set to 0. In our proposed model, we set the number of hidden factors $M = 32$, the dimension of the feature embedding $H = 32$, the number of rnn layers $L = 2$. In contrastive learning, we set the temperature parameter $\tau = 0.1$ and the balance hyperparameter $\gamma = 0.1$.

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Baselines

The following are the details of the baselines used in our experiments:

- **MLP**: A multi-layer perceptron model with a linear prediction layer.
- **GRU**: A neural network consisting of a gated recurrent unit layer followed by a linear prediction layer.
- **TCN**: A temporal convolutional network evaluated empirically for sequence modeling tasks.
- **Transformer**: A neural network employing the Transformer architecture for sequence prediction.
- **ALSTM**: A dual-stage attention-based recurrent neural network designed for time series prediction, featuring a temporal attention aggregation layer built upon a standard LSTM.
- **SFM**: A recurrent neural network utilizing discrete Fourier transform to decompose the hidden states of memory cells, enabling the capture of multi-frequency trading patterns from historical market data for stock price prediction.
- **GAT**: A graph attention network that applies masked self-attention layers to graph-structured data. The nodes in the stacked layers have varying weights and can attend to their neighbors' features without requiring costly matrix operations (e.g., inversion) or prior knowledge of the graph structure.
- **HIST**: A graph-based model with a doubly residual architecture designed to uncover hidden concepts for stock trend forecasting.
- **HyperGCN**: A hypergraph convolutional neural network that extends the capabilities of traditional graph convolutional networks by operating on hypergraphs, which offer greater expressiveness.
- **STHAN-SR**: A neural spatiotemporal hypergraph model for stock selection that integrates a ranking loss function

Methods	$\Delta t = 1$		$\Delta t = 5$		$\Delta t = 10$		$\Delta t = 20$	
	MSE	F1	MSE	F1	MSE	F1	MSE	F1
MLP	0.976	**0.519**	0.995	0.512	0.999	0.512	0.998	0.512
GRU	0.979	0.494	0.997	0.492	1.002	0.494	1.003	0.490
TCN	0.977	0.485	0.993	0.485	0.996	0.487	0.996	0.489
Transformer	0.978	0.474	0.995	0.467	0.998	0.468	0.999	0.474
ALSTM	0.979	0.500	0.995	0.504	0.998	0.506	0.999	0.506
SFM	0.976	0.507	0.992	0.503	0.993	0.505	0.993	0.507
GAT	0.976	0.468	0.992	0.464	0.995	0.465	0.996	0.468
HIST	0.975	0.476	0.990	0.491	0.993	0.498	0.997	0.500
HyperGCN	0.980	0.504	1.002	0.500	1.005	0.503	1.004	0.506
STHAN-SR	0.977	0.492	0.994	0.492	0.996	0.488	0.996	0.487
CI-STHPAN	0.981	0.473	1.003	0.450	1.009	0.439	1.013	0.421
FactorVAE	0.980	0.433	0.991	0.446	0.997	0.411	1.001	0.474
-woPrior	0.977	0.511	0.995	0.510	0.998	0.515	0.997	0.519
-woHidden	0.975	0.504	0.991	0.504	0.994	0.506	0.995	0.508
-woAlpha	0.977	0.507	0.995	0.516	0.997	0.520	0.996	0.517
-woCL	0.976	0.510	0.991	0.509	0.994	0.514	0.996	0.515
FactorGCL	**0.973**	**0.519**	**0.989**	**0.531**	**0.991**	**0.528**	**0.992**	**0.535**

Table 1: The performance comparison of different models on stock returns prediction. The MSE metric is the lower the better, and the F1 score is the higher the better.

and a self-attention mechanism to capture spatiotemporal dependencies in stock data.

- **FactorVAE**: A probabilistic factor model based on a variational autoencoder that extracts latent factors from noisy data to predict cross-sectional stock returns.
- **CI-STHPAN**: A hypergraph-based pre-trained Transformer model for stock ranking, incorporating a channel-independent spatio-temporal hypergraph attention network.

Evaluation Metrics

In our experiments, we use the IC and ICIR metrics to evaluate the performance of the stock returns prediction models. The IC metric measures the correlation between the predicted stock returns and the actual stock returns, while the ICIR metric is the IC divided by the standard deviation of the IC, to reflect the stability of the prediction performance. The higher the IC and ICIR values, the better the prediction performance of the model. Formally, the IC and ICIR are defined as follows:

$$IC_t = \frac{1}{N} \frac{\left(\hat{y}_t - \text{mean}\left(\hat{y}_t\right)\right)^T \left(y_t - \text{mean}\left(y_t\right)\right)}{\text{std}\left(\hat{y}_t\right) \cdot \text{std}\left(y_t\right)}$$

$$IC = \text{mean}(IC_t)$$

$$ICIR = \frac{\text{mean}(IC_t)}{\text{std}(IC_t)}$$

We alse report MSE and F1 score as additional metrics. The MSE metric measures the mean squared error of regression between the predicted stock returns and the actual stock returns, while the F1 score is the harmonic mean of precision and recall, to evaluate the classification performance of the model. The results of the additional metrics are shown in Table 1. We can see that our proposed model outperforms the baselines in terms of MSE and F1 score, which further demonstrates the effectiveness of our model in stock returns prediction.

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} \left(y_i - \hat{y}_i\right)^2$$

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

where TP, FP, and FN are the number of true positive, false positive, and false negative samples, respectively, they are caclulated by the following rules:

$$\text{TP} = \sum_{i=1}^{N} \mathbb{I}\left(y_i > 0 \wedge \hat{y}_i > 0\right)$$

$$\text{FP} = \sum_{i=1}^{N} \mathbb{I}\left(y_i < 0 \wedge \hat{y}_i > 0\right)$$

$$\text{FN} = \sum_{i=1}^{N} \mathbb{I}\left(y_i > 0 \wedge \hat{y}_i < 0\right)$$

In the simulation investment experiment, we plot the cumulative return (CR) of the investment strategy over time to evaluate the performance of the compared models, and use the annualized return (AR), information ratio(IR), and return over maximum drawdown (RoMaD) of cumulative excess return (CER) to evaluate the performance of the stock returns prediction models. The AR metric measures the profitability of the investment strategy, the IR metric measures

Methods	$\Delta t = 1$		$\Delta t = 5$		$\Delta t = 10$		$\Delta t = 20$	
	IC	ICIR	IC	ICIR	IC	ICIR	IC	ICIR
MLP	0.0154	0.3381	0.0324	0.6089	0.0428	0.7561	0.0560	1.0903
GRU	0.0288	0.5253	0.0363	0.6300	0.0466	0.7821	0.0675	1.1898
TCN	0.0326	0.4442	0.0369	0.6005	0.0496	0.8504	0.0677	1.2004
Transformer	0.0194	0.4500	0.0368	0.7174	0.0468	0.8823	0.0636	1.2004
ALSTM	0.0158	0.3602	0.0267	0.4870	0.0348	0.6836	0.0539	1.1251
SFM	0.0340	0.5697	0.0459	0.8029	0.0572	0.9362	0.0713	1.1888
GAT	0.0193	0.4167	0.0327	0.6530	0.0421	0.8112	0.0552	1.0758
HIST	0.0187	0.4107	0.0325	0.6713	0.0425	0.8649	0.0566	1.1454
HyperGCN	0.0402	0.3237	0.0349	0.4177	0.0439	0.5593	0.0605	0.9015
STHAN-SR	0.0151	0.2938	0.0288	0.4653	0.0407	0.6079	0.0585	0.9520
FactorVAE	0.0253	0.4741	0.0360	0.7018	0.0609	**1.0944**	0.0578	0.9485
CI-STHPAN	0.0172	0.3706	0.0279	0.5238	0.0389	0.7198	0.0590	1.1297
FactorGCL	**0.0405**	**0.6978**	**0.0511**	**0.8196**	**0.0646**	0.9911	**0.0902**	**1.4669**

Table 2: The stock returns prediction performance of all compared methods on the Hong Kong stock market.

the risk-adjusted return, and the RoMaD metric measures the return relative to the maximum drawdown. The higher the AR, IR, and RoMaD values, the better the investment performance of the model. Formally, the CR, CER, AR, IR, and RoMaD are defined as follows:

$$\text{CR}^{(t)} = \sum_{i=1}^{t} r^{(i)}$$

where $r^{(i)}$ is the return at time i.

$$\text{CER}^{(t)} = \sum_{i=1}^{t} \left(r^{(i)} - r_b^{(i)} \right)$$

where $r^{(i)}$ is the return of the strategy at time i, and $r_b^{(i)}$ is the return of benchmark.

$$\text{AR} = \frac{\text{CER}^{(T_{\text{test}})}}{T} \times 252$$

where T_{test} is the total number of trading days in the test set, and 252 is the typical number of trading days in a year.

$$\text{IR} = \frac{\mathbb{E}\left[r^{(i)} - r_b^{(i)} \right]}{\sigma_{r^{(i)} - r_b^{(i)}}} \times \sqrt{252}$$

where $r^{(i)}$ is the return of the strategy at time i, $r_b^{(i)}$ is the return of the benchmark at time i, and $\sigma_{r^{(i)} - r_b^{(i)}}$ is the standard deviation of the excess returns.

$$\text{RoMaD} = \frac{\text{AR}}{\text{Max Drawdown}}$$

where Max Drawdown is defined as the maximum observed loss from a peak to a trough of a portfolio, before a new peak is attained.

More Experiment Results

To evaluate the generalization ability of our proposed model, we conduct experiments on different stock market. Specifically, we performe experiments on the Hong Kong stock market, which included 1834 stocks. The market data and the experimental settings are the same as the previous experiments. The results are shown in Table 2. We can see that our proposed model outperforms the baselines on the Hong Kong stock market, which demonstrates the generalization ability of our model on different stock markets.

Methods	AR	IR	RoMaD
MLP	-0.003	-0.048	-0.009
GRU	0.117	1.928	0.595
TCN	0.040	0.586	0.153
Transformer	0.078	1.235	0.501
ALSTM	0.077	1.432	0.562
SFM	0.073	1.148	0.392
GAT	0.005	0.084	0.018
HIST	-0.031	-0.507	-0.095
HyperGCN	0.021	0.312	0.084
STHAN-SR	0.101	1.660	0.678
FactorVAE	0.044	0.681	0.217
CI-STHPAN	0.058	0.937	0.182
FactorGCL	**0.127**	**1.995**	**1.256**

Table 3: The investment simulation results of all compared methods on the Hong Kong stock market.

Additionally, we alse we conducted a simulation backtest experiment using the Hang Seng Index stocks as the benchmark, which consists of the most representative stocks in the Hong Kong stock market. We employed the same trading strategy as described in the paper, with *TopK* $= 10$, $\Delta t = 10$, and the transaction cost to 0.3%. The results are shown in Table 3.